Hotel Booking Demand & Cancellation Analysis

Case Study Report
for the course
Big Data Analytics - 20CS41001

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By

B. Jaipreeth – 23R11A0506
S. Mohana Krishna – 23R11A0534
V. Jaya Prakash – 23R11A0544

Under the guidance of
P Usha Shree

Department of Computer Science and Engineering
Accredited by NBA

Geethanjali College of Engineering and Technology

(UGC Autonomous)
(Affiliated to J.N.T.U.H, Approved by AICTE, New Delhi)
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1. Introduction

Across the 119,390 bookings analysed in this case study, 37.0% were cancelled. Those cancellations left about 16.7 million of potential room revenue unrealised against 26.0 million actually earned. Cancellations are therefore the central business problem addressed by this report.

1.1 Problem Context

A hotel sells a perishable product. A room that stays empty tonight cannot be sold tomorrow, so every cancelled booking is revenue that is lost unless the room is resold at short notice. Hotel managers work under three linked pressures:

- **Demand is seasonal.** Occupancy and room rates rise and fall through the year, and the pattern differs between a business-oriented city hotel and a leisure-oriented resort.
- **Bookings are uncertain.** Guests book months ahead through many channels – online travel agents, tour operators, corporate accounts and direct enquiries – and these channels cancel at very different rates.
- **Revenue is concentrated.** A small number of source markets and booking channels generate most of the income, and also most of the cancellations.

The dataset used here describes bookings at two hotels in Portugal, a City Hotel and a Resort Hotel, with arrival dates from July 2015 to August 2017. Managers of such properties need to know when demand peaks, which guests cancel, how far ahead cancellations happen and where the money is lost. A spreadsheet of raw booking rows cannot answer these questions at a glance.

1.2 Objective of the Analysis

The objective is to build an interactive Tableau dashboard that explains hotel demand and cancellation behaviour. The analysis is guided by five questions:

1. **Overview:** How large is the business, and how much revenue is realised against how much is lost to cancellations?
2. **Hotel comparison:** How do the City Hotel and the Resort Hotel differ in volume, price and cancellation behaviour?
3. **Temporal analysis:** How do realised revenue and average daily rate (ADR) vary across the months of the year?
4. **Risk analysis:** How do lead time, market segment, customer type and deposit type relate to the probability that a booking is cancelled?
5. **Spatial analysis:** Which guest countries produce the most revenue, and which of them cancel the most?

1.3 How Tableau Addresses the Problem

Tableau was selected because the data is large (119,390 rows and 32 columns) and the questions above have to be read together rather than one at a time. Tableau is used in this project to:

- derive the business measures the raw file lacks – total nights, estimated revenue, realised revenue, lost revenue, cancellation rate and lead-time bracket – as calculated fields, without altering the source CSV;
- show geography with a symbol map, seasonality with line charts, and category comparisons with bar charts, each chart type matched to the question it answers;

- combine all the views into a single dashboard where a manager can compare hotels, months and segments side by side, with tooltips giving exact values on demand.

The rest of the report covers the dataset (Section 2), the methodology followed in Tableau (Section 3), the dashboard design with figures (Section 4), the analysis and insights (Section 5) and the conclusions (Section 6).

2. Dataset Description

The dataset is a single CSV file of 119,390 hotel bookings described by 32 attributes, covering two hotels over arrival dates from July 2015 to August 2017.

2.1 Dataset Name and Size

Property	Value
File name	hotel_booking2.csv
Subject	Hotel booking demand (City Hotel and Resort Hotel)
File size	Approximately 16.9 MB
Rows (bookings)	119,390
Columns (attributes)	32 (16 integer, 4 decimal, 12 text)
City Hotel bookings	79,330 (66.4%)
Resort Hotel bookings	40,060 (33.6%)
Arrival years covered	2015, 2016, 2017
Distinct guest countries	177
Market segments	8

Each row represents one booking, which was either checked out, cancelled or recorded as a no-show.

2.2 Attributes and Their Meaning

The attributes used directly in the dashboard are listed first, followed by the remaining attributes that describe the booking.

Attribute	Meaning
hotel	Hotel type: City Hotel or Resort Hotel
is_canceled	1 if the booking was cancelled, 0 otherwise
lead_time	Number of days between the booking date and the arrival date
arrival_date_year / month / week_number / day_of_month	The arrival date, split into its parts
stays_in_weekend_nights / stays_in_week_nights	Number of weekend and weekday nights booked
adr	Average daily rate: total lodging cost divided by the number of nights
country	Guest's country of origin, as an ISO three-letter code
market_segment	Booking segment, e.g. Online TA, Offline TA/TO, Groups, Direct, Corporate
distribution_channel	Channel used for the booking: TA/TO, Direct, Corporate or GDS
customer_type	Contract, Group, Transient or Transient-Party
deposit_type	No Deposit, Non Refund or Refundable
adults / children / babies	Number of guests in each category
meal	Meal plan booked (BB, HB, FB or SC)

Attribute	Meaning
is_repeated_guest	1 if the guest has stayed at the property before
previous_cancellations / previous_bookings_not_canceled	The guest's earlier cancelled and honoured bookings
reserved_room_type / assigned_room_type	Room type requested and room type actually given
booking_changes	Number of changes made to the booking after it was created
agent / company	Identifier of the travel agent or company that made the booking
days_in_waiting_list	Days the booking waited before it was confirmed
required_car_parking_spaces	Number of parking spaces requested
total_of_special_requests	Number of special requests made by the guest
reservation_status / reservation_status_date	Final status (Check-Out, Canceled, No-Show) and the date it was set

2.3 Cleaning and Preprocessing Steps

The CSV was connected directly to Tableau Desktop and the source file was left unmodified. The table below records what the data quality checks found and how each issue was handled.

Issue found	Size	Treatment
Missing country values	488 rows	Left as Null; these bookings do not appear on the Regional Map
Missing children values	4 rows	Left as Null; children is not used in any view
agent and company stored as the text "NULL"	16,340 and 112,593 rows	Not used in any view, so no cleaning was required
Zero or negative adr	1,960 rows (one negative value, –6.38)	Kept; zero-priced bookings contribute zero revenue and the single negative value is negligible
Extreme adr value	Maximum of 5,400	Kept; it slightly raises the average, so average ADR is read together with the revenue totals
Bookings with zero nights	715 rows	Kept; their estimated revenue evaluates to zero
Exact duplicate rows	31,994 rows	Kept; the file has no booking identifier, so identical rows cannot be proved to be repeats
Text-coded geography	country holds ISO codes	Assigned the Country geographic role in Tableau so the map can plot them
Month names stored as text	arrival_date_month	Sorted manually into calendar order on the Monthly Seasonality view

The substantive preparation work took place inside Tableau in the form of calculated fields, which are described in Section 3.2.

3. Methodology / Approach

The work followed four stages: importing the CSV, deriving business measures as calculated fields, building seven worksheets, and assembling those worksheets into a single dashboard.

3.1 Importing the Data

The single file hotel_booking2.csv was connected to Tableau Desktop as a text-file data source. Because the whole analysis draws on one table, no joins, unions or data blending were required. The Country geographic role was applied to the country field so that Tableau could place each guest country on a map, and the numeric fields were confirmed as measures while the descriptive fields were confirmed as dimensions.

3.2 Data Preparation: Calculated Fields

The raw file contains no revenue column and no cancellation-rate column, so six calculated fields were created to support the dashboard.

Field	Formula	Purpose
cf_Total_Nights	[stays_in_weekend_nights] + [stays_in_week_nights]	Length of stay in nights
cf_Estimated_Revenue	[adr] * [cf_Total_Nights]	Value of a booking if it is honoured
cf_Realized_Revenue	IF [is_canceled] = 0 THEN [cf_Estimated_Revenue] ELSE 0 END	Revenue from stays that actually took place
cf_Lost_Revenue	IF [is_canceled] = 1 THEN [cf_Estimated_Revenue] ELSE 0 END	Revenue forgone through cancellations
cf_Cancellation_Rate	SUM([is_canceled]) / COUNT([hotel])	Share of bookings that were cancelled
cf_Lead_Time_Bracket	IF [lead_time] <= 7 THEN "1. 0–7 Days" ELSEIF [lead_time] <= 30 THEN "2. 8–30 Days" … ELSE "6. 365+ Days" END	Groups lead time into six ordered brackets: 0–7, 8–30, 31–90, 91–180, 181–365 and 365+ days

The lead-time brackets were prefixed "1." to "6." so that Tableau sorts them in time order rather than alphabetically. Estimated revenue is calculated as ADR multiplied by nights, not taken from billing records, so every revenue figure in this report is a model value rather than an accounting figure.

3.3 Chart Selection Rationale

Each worksheet answers one question, and the chart type was chosen to suit that question.

Worksheet	Chart type	Question answered	Why this chart
KPI Cards	Text cards (four measures)	How large is the business?	Headline numbers give scale before any detail is read
Hotel Split	Pie chart	What share of bookings does each hotel hold?	Only two parts, so a pie is read instantly
Monthly Seasonality	Dual line chart, one line per hotel	How do revenue and ADR change by month?	Lines show change across an ordered sequence of months

Worksheet	Chart type	Question answered	Why this chart
Lead Time Risk	Horizontal bars coloured by rate	How does cancellation rise with lead time?	Bars compare ordered brackets; colour reinforces the rate
Market Segment	Horizontal bars with rate labels	Which segments bring volume, and which cancel?	Long segment names fit comfortably on a horizontal axis
Customer Insights	Stacked bars by deposit type	How do customer and deposit type affect cancellation?	Colour splits each bar into the deposit types being compared
Regional Map	Symbol map	Where do the guests come from?	Country is geographic; size and colour add two measures

3.4 Dashboard Design Process

Layout. All seven worksheets were combined into one dashboard, titled "Hotel Booking Demand & Cancellation Analysis". The four KPI cards form a header band across the top so that the reader sees the scale of the business first. Below them the dashboard is laid out in two rows: Monthly Seasonality, Regional Map, Lead Time Risk and Market Segment on the upper row, and Customer Insights and Hotel Split on the lower row. The dashboard uses a tiled layout so that every chart resizes cleanly, and the shared colour legends sit on the right-hand edge.

Interactivity. Each view carries a hover tooltip that reveals its exact values, and the colour legends on the right can be used to highlight a category across the dashboard. The map supports pan and zoom for the crowded European markets. The current workbook does not define filter actions or parameters; adding them is listed as future work in Section 6.3.

Colour scheme. Colour was applied deliberately rather than decoratively:

- hotel type is given one colour per hotel – blue for the City Hotel and orange for the Resort Hotel – and the same pair is reused in the pie chart and the line chart so the two properties are always recognisable;
- cancellation rate, which is a risk measure, is encoded as a continuous blue gradient in the Lead Time Risk and Market Segment bars and on the Regional Map, so that a darker mark always means a higher risk;
- deposit type uses three distinct hues in the Customer Insights view, because the three deposit types have no natural order and should not be read as a scale.

4. Dashboard Design

The final analytical product is one dashboard built from seven worksheets, moving from headline numbers to detailed risk and geography. Figure 1 shows the complete dashboard, and Figures 2 to 8 show each worksheet in turn.

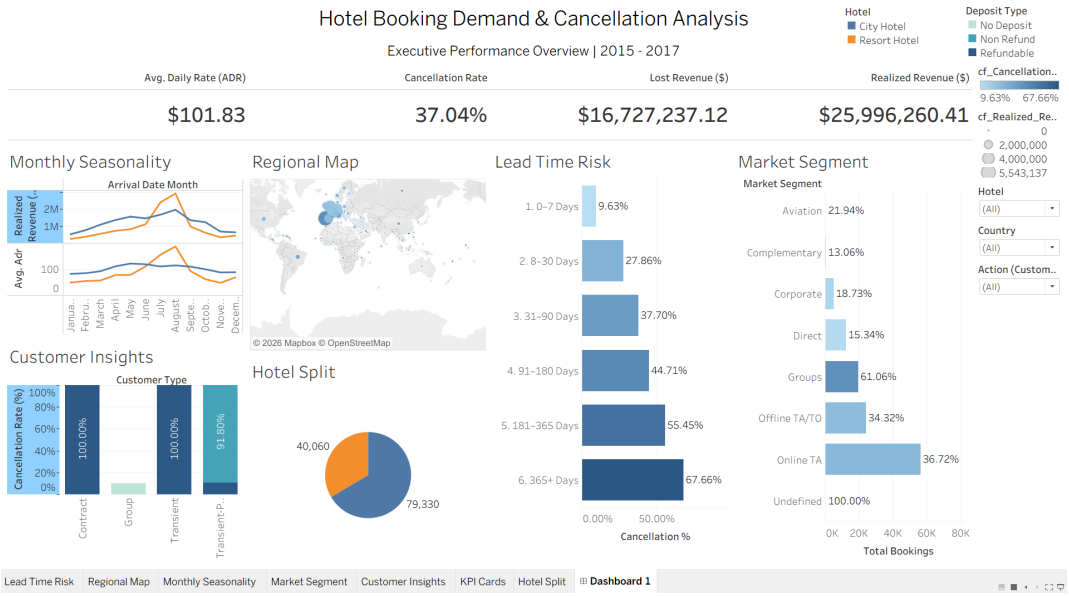


Figure 1: *Dashboard 1 – the complete Hotel Booking Demand & Cancellation Analysis dashboard, with the KPI band at the top, the seasonality, map, lead-time and segment views in the middle, and the customer and hotel views below.*

As Figure 1 shows, the dashboard is organised into three zones. The KPI band answers "how big and how serious is the problem", the middle row answers "when, where and who", and the bottom row answers "which customers and which property". The sections below explain each worksheet in the order a reader would use them.

4.1 Overview: KPI Cards and Hotel Split

KPI Cards

Figure 2 presents four headline measures: average ADR, cancellation rate, lost revenue and realised revenue. Together they state the business problem in numbers – how much is earned, how much is lost, and what share of bookings is cancelled. Text cards were chosen because a single number needs no axis, and the reader gains the scale of the data before interpreting any chart.

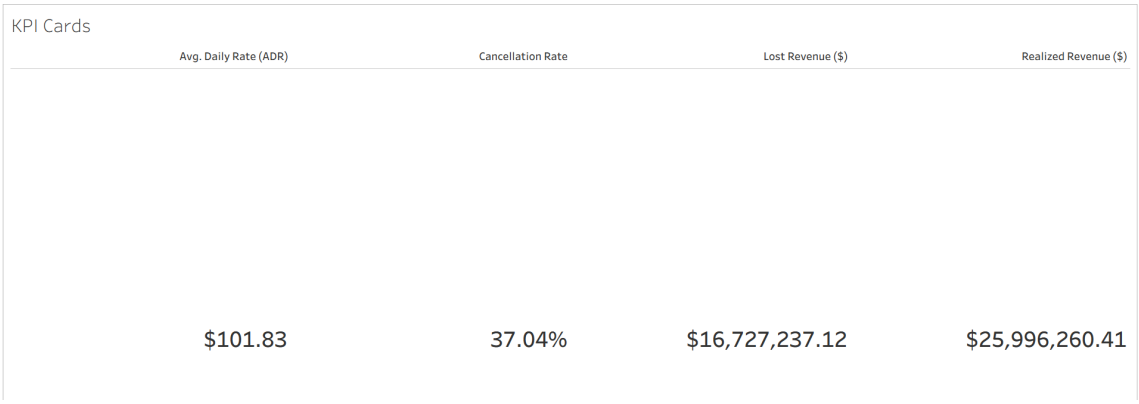


Figure 2: *KPI Cards worksheet showing average ADR of 102, a cancellation rate of 37.04%, lost revenue of 16,727,237 and realised revenue of 25,996,260.*

Hotel Split

Figure 3 divides the bookings between the City Hotel and the Resort Hotel. This matters because every other measure on the dashboard differs by property, so the reader needs to know the mix before reading the other views. A pie chart suits the comparison because there are only two parts and the reader only needs to see which is larger and roughly by how much.

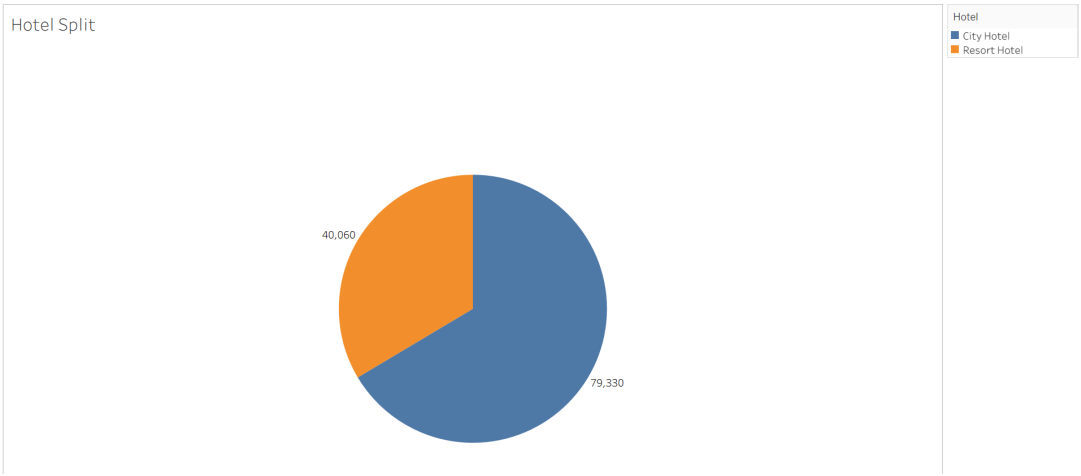


Figure 3: *Hotel Split worksheet – 79,330 City Hotel bookings against 40,060 Resort Hotel bookings.*

4.2 Temporal and Risk Views

Monthly Seasonality

Figure 4 plots realised revenue in the upper panel and average ADR in the lower panel, by arrival month, with one line for each hotel. It answers when each property earns its money and whether higher prices coincide with higher revenue. Lines were chosen because the months form an ordered sequence; a line makes the peaks, the troughs and the gap between the two hotels visible at a glance, which a bar chart would fragment.

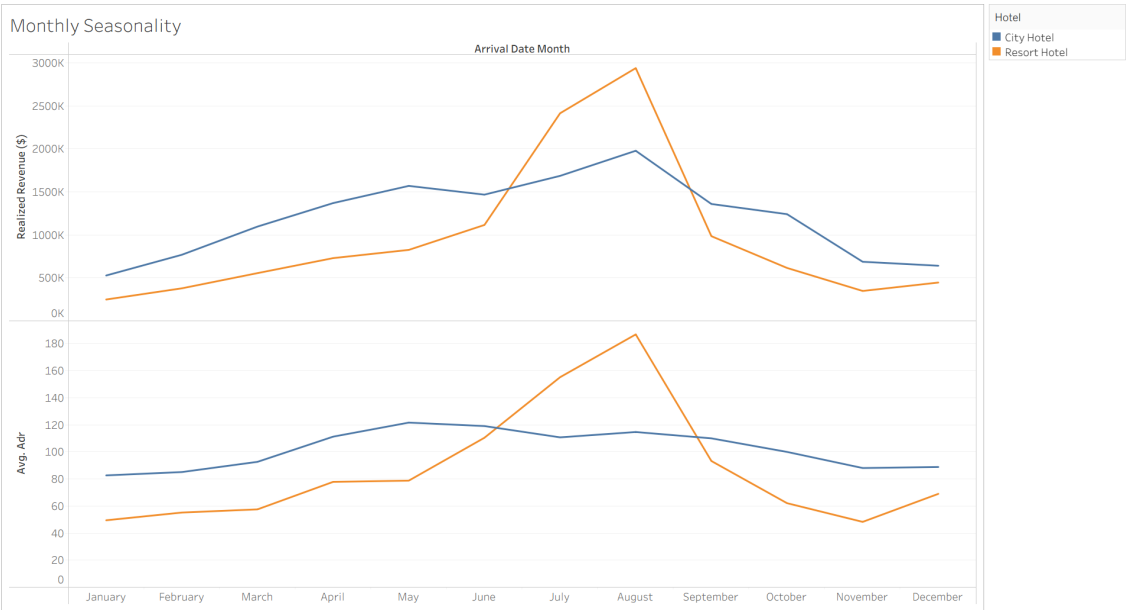


Figure 4: *Monthly Seasonality worksheet – realised revenue (upper panel) and average ADR (lower panel) by arrival month for each hotel.*

The orange Resort Hotel line peaks sharply in July and August in both panels, while the blue City Hotel line is noticeably flatter, which is the clearest single contrast between the two properties.

Lead Time Risk

Figure 5 shows the cancellation rate for each of the six lead-time brackets, with the bars coloured by the rate itself. It tests whether guests who book further ahead are more likely to cancel. Horizontal bars keep the brackets in time order from top to bottom, and the colour gradient makes the riskiest brackets stand out even before the labels are read.

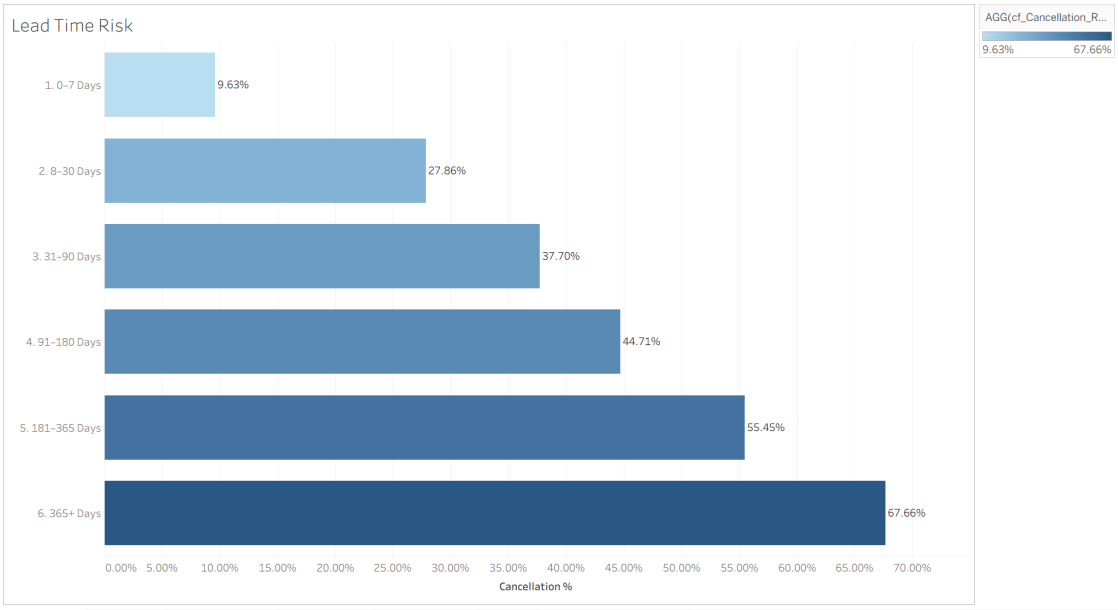


Figure 5: *Lead Time Risk worksheet – cancellation rate by lead-time bracket, rising from 9.63% to 67.66%.*

4.3 Segment and Customer Views

Market Segment

Figure 6 counts the bookings in each market segment and labels every bar with that segment's cancellation rate, while the bar colour also encodes the rate. Pairing volume with risk is exactly what a revenue manager needs, because a large segment with a high rate costs far more than a small one. Horizontal bars were chosen to accommodate long segment names such as "Offline TA/TO" and "Complementary".

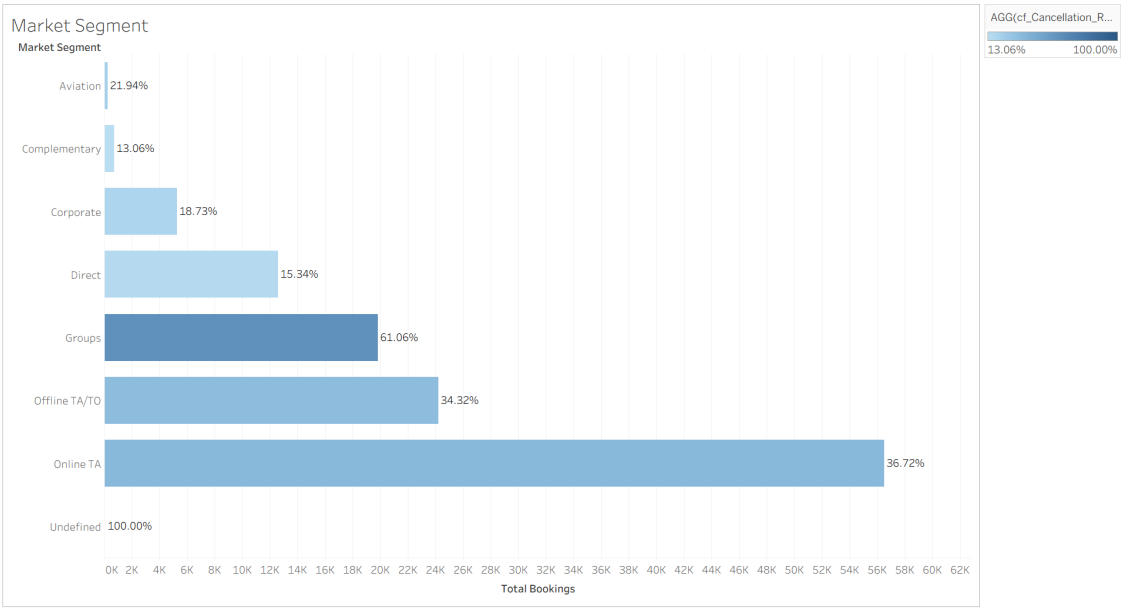


Figure 6: *Market Segment worksheet – booking count by segment, labelled and coloured by cancellation rate.*

Customer Insights

Figure 7 shows the cancellation rate for each customer type, with each bar split by deposit type. It checks whether requiring a deposit changes cancellation behaviour within each customer group. Colour carries the deposit type so that the three types can be compared inside every customer group. Because the bars stack the rate of each deposit type rather than a share of a whole, the axis runs beyond 100%; the segment labels, not the bar heights, should be read as the result.

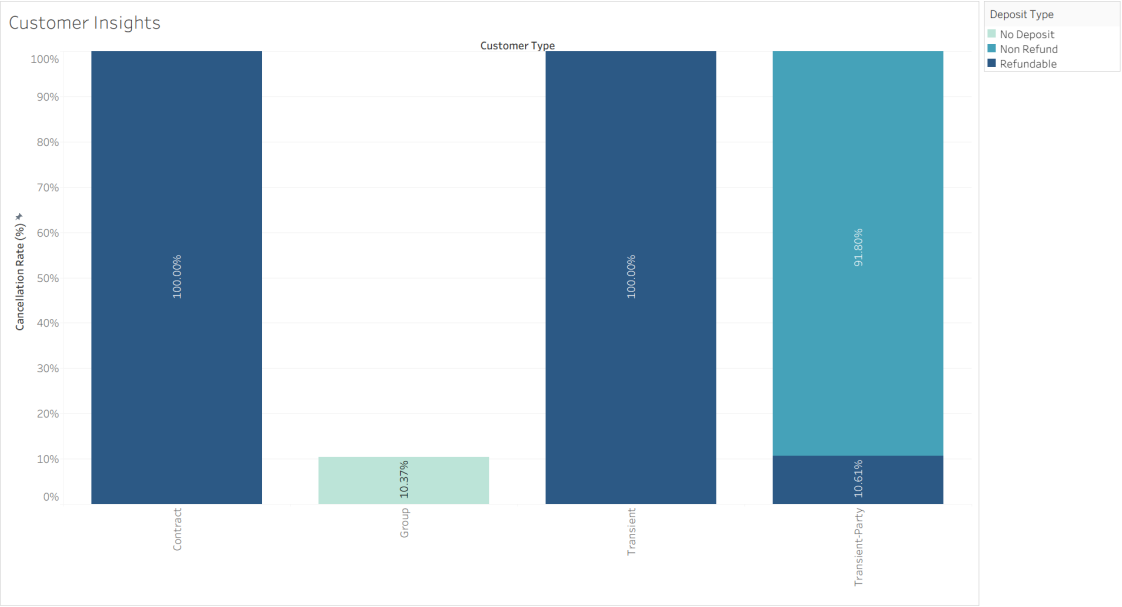


Figure 7: *Customer Insights worksheet – cancellation rate by customer type, split by deposit type.*

The orange Non Refund segments reach 100% for Contract and Transient customers and 91.80% for Transient-Party, which is the anomaly examined in Section 5.6.

4.4 Regional Performance Map

Figure 8 is a symbol map carrying one circle for each guest country. Circle size encodes realised revenue and circle colour encodes cancellation rate, so a large dark circle marks a country that brings a great deal of revenue and also cancels frequently. A map was chosen because country is a geographic attribute and the shape of the source markets – heavily European, with a thin scattering worldwide – is grasped far faster spatially than from a ranked list.

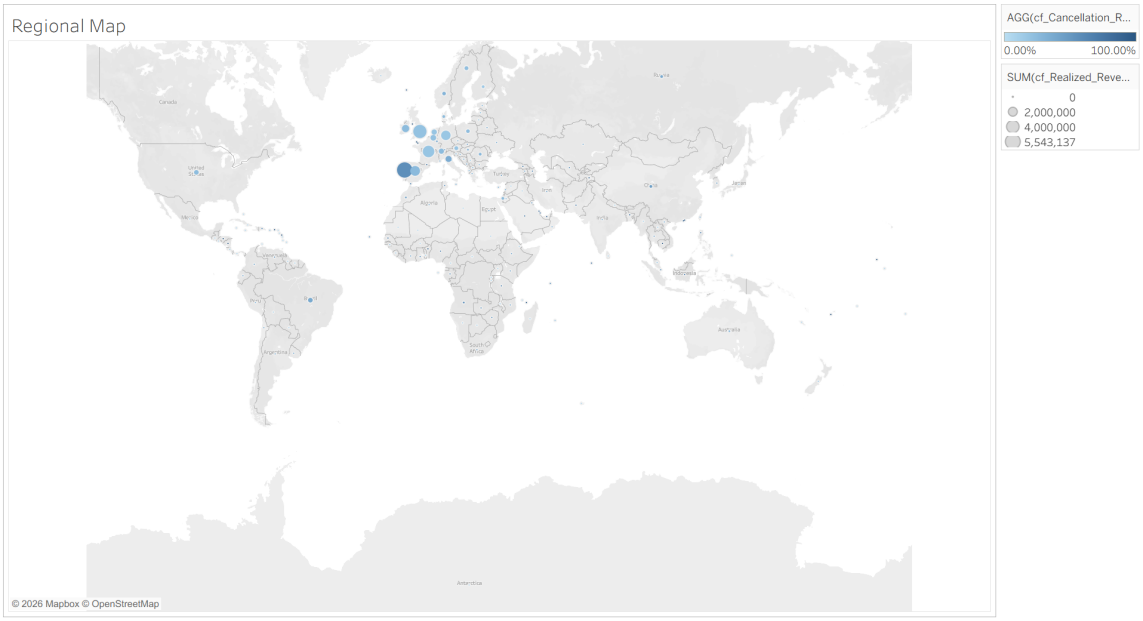


Figure 8: *Regional Map worksheet – guest countries sized by realised revenue and coloured by cancellation rate.*

The largest and darkest circle sits on Portugal, which combines the highest revenue with the highest cancellation rate of the major markets. Three records could not be mapped, as the "3 unknown" indicator in the lower-right corner of the view reports.

4.5 Dashboard Interactivity

Feature	Where it is used	Benefit to the reader
Hover tooltips	All seven worksheets	Exact values appear on demand without cluttering the charts
Colour legends	Hotel, cancellation rate and deposit type legends	Selecting a legend entry highlights that category
Data labels	Lead Time, Market Segment and Customer Insights bars	Rates can be read directly without hovering
Map pan and zoom	Regional Map	Allows close inspection of the crowded European markets
Tiled dashboard layout	Dashboard 1	Every chart resizes cleanly and stays legible on one screen

5. Analysis and Insights

Read together, the eight views show that cancellations are concentrated rather than random: they are driven by a few countries, a few segments, one booking habit and one unusual deposit type. Each subsection below reports the finding from one view, and every figure quoted is calculated from the full dataset and matches the corresponding Tableau worksheet.

5.1 Overall Business Health (Figure 2)

The 119,390 bookings carry an average ADR of 101.83 and an overall cancellation rate of 37.04%. Estimated revenue across all bookings is 42.72 million, of which 26.00 million (60.8%) was realised and 16.73 million (39.2%) was lost to cancellations. The lost share of revenue is slightly larger than the cancelled share of bookings, which means that cancelled bookings were on average a little more valuable than honoured ones.

5.2 Hotel Comparison (Figures 1 and 3)

The City Hotel holds two thirds of all bookings, but the Resort Hotel earns far more per booking. The two properties differ on every measure:

Measure	City Hotel	Resort Hotel
Bookings	79,330	40,060
Cancellation rate	41.7%	27.8%
Average ADR	105.30	94.95
Average nights per booking	2.98	4.32
Average lead time (days)	109.7	92.7
Realised revenue	14.39 million	11.60 million
Lost revenue	10.89 million	5.84 million

The City Hotel has twice the booking volume but only 24% more realised revenue than the Resort, and it loses almost twice as much to cancellations. Guests at the Resort stay about 1.3 nights longer and cancel far less often, so each Resort booking is both more valuable and more reliable.

5.3 Seasonality (Figure 4)

Seasonality is strong at the Resort Hotel and mild at the City Hotel.

- **Resort Hotel:** realised revenue peaks in August at 2.94 million and is lowest in January at 0.25 million, a swing of almost twelve times. July and August together account for 46% of all Resort revenue. Average ADR follows the same shape, rising from 49.5 in January to 186.8 in August, so the peak is driven by both more guests and higher prices.
- **City Hotel:** revenue is far steadier. It rises from 0.53 million in January to a broad plateau of 1.4 to 2.0 million between April and August, with the highest month again August at 1.98 million. ADR varies much less, from 82.6 in January to 121.6 in May.

A caveat on the monthly view. The data covers July 2015 to August 2017, so July and August contain three years of arrivals while every other month contains only two. This inflates the July and August totals by roughly half. The summer peak is still genuine, because the Resort's ADR also peaks in those months, but the size of the peak in Figure 4 should be read with this in mind.

5.4 Lead Time and Cancellation Risk (Figure 5)

The cancellation rate climbs steadily with how far ahead the guest booked.

Lead-time bracket	Bookings	Cancellation rate
1. 0–7 days	19,746	9.63%
2. 8–30 days	18,960	27.86%
3. 31–90 days	29,553	37.70%
4. 91–180 days	26,439	44.71%
5. 181–365 days	21,544	55.45%
6. 365+ days	3,148	67.66%

Bookings made within the last week are almost always honoured, whereas those made more than a year ahead are cancelled about two times in three – a sevenfold difference. The longer the gap between booking and arrival, the more chance that plans, prices or alternatives change. Bookings made more than 90 days ahead account for 43% of all bookings and carry most of the risk.

5.5 Market Segment (Figure 6)

Segment	Bookings	Cancellation rate	Average ADR
Online TA	56,477	36.72%	117.20
Offline TA/TO	24,219	34.32%	87.35
Groups	19,811	61.06%	79.48
Direct	12,606	15.34%	115.45
Corporate	5,295	18.73%	69.36
Complementary	743	13.06%	2.89
Aviation	237	21.94%	100.14

Online travel agents supply almost half of all bookings (47.3%) with a cancellation rate close to the overall average. Groups are the riskiest segment by a wide margin: 61.06% of group bookings are cancelled and they pay a below-average ADR of 79.48, so they are both low in value and unreliable. Direct bookings are the opposite, with a cancellation rate of 15.34% and an ADR almost as high as the online agents'. Direct and corporate business are therefore the safest sources of revenue. The "Undefined" bar at 100% in Figure 6 represents only two records and can be disregarded.

5.6 Customer Type and Deposit Type (Figure 7)

By customer type, Transient guests cancel most often at 40.7%, followed by Contract at 31.0%, Transient-Party at 25.4% and Group at 10.2%.

The deposit result is counter-intuitive. Bookings carrying a Non Refund deposit are cancelled 99.4% of the time across 14,587 bookings, compared with 28.4% for No Deposit and 22.2% for Refundable. A non-refundable rule would be expected to discourage cancellation, so this almost certainly reflects how such bookings are recorded rather than real guest behaviour. Of the Non Refund bookings, 9,172 fall in the Groups segment and 97% come from Portugal, so a small number of large Portuguese group or agency blocks probably explain the pattern. These records should be investigated before any judgement is made about deposit policy.

5.7 Regional Performance (Figure 8)

Country	Bookings	Realised revenue	Cancellation rate
Portugal	48,590	5.54 million	56.6%
United Kingdom	12,129	4.11 million	20.2%
France	10,415	3.10 million	18.6%
Spain	8,568	2.25 million	25.4%
Germany	7,287	2.07 million	16.7%

These five countries together produce 65.9% of realised revenue. Portugal is the largest source market, with 40.7% of all bookings, but it also cancels at 56.6% against 23.6% for all other countries combined. Portuguese guests therefore account for 62% of every cancellation in the dataset and about 8.59 million of the 16.73 million lost revenue, which is 51% of the total loss. The United Kingdom, by contrast, brings 4.11 million of realised revenue from fewer than a quarter of Portugal's bookings, precisely because its guests rarely cancel.

5.8 Answers to the Objective Questions

Question from Section 1.2	Answer from the dashboard
How much revenue is lost?	16.73 million, which is 39.2% of estimated revenue
How do the two hotels differ?	The City Hotel has the volume and the high cancellations; the Resort has longer stays, lower cancellations and a sharp summer peak
When is revenue earned?	Mainly from April to September; the Resort peaks sharply in July and August
What drives cancellation?	Long lead times, the Groups and Online TA segments, and Portuguese guests
Where does the money come from?	Portugal, the United Kingdom, France, Spain and Germany, which supply two thirds of realised revenue

6. Conclusions

The two hotels lose 39.2% of their potential room revenue to cancellations, and that loss comes mainly from long-lead bookings, group and agency business, and Portuguese guests. A fall of just one percentage point in the overall cancellation rate would recover roughly 0.45 million, assuming the cancelled bookings are of average value, which makes this a worthwhile target for management attention.

6.1 Actionable Insights and Recommendations

Finding from the dashboard	Recommended action
Bookings made 181 days or more ahead (20.7% of bookings) are cancelled 55% to 68% of the time	Ask for a partial prepayment on early bookings, send a reconfirmation request 30 to 60 days before arrival, and shorten the free-cancellation window
Groups are cancelled 61.06% of the time at a below-average ADR	Add attrition clauses, rooming-list deadlines and staged deposits to group contracts, and hold fewer rooms in group blocks
Portugal supplies 62% of all cancellations and about half of the lost revenue	Review the Portuguese agencies and group blocks behind these bookings and apply stricter terms to them
Direct bookings cancel at 15.34% against 36.72% for Online TA, at a similar ADR	Offer direct-booking incentives such as a best-rate guarantee or a loyalty benefit to shift volume away from online agents
The Resort earns 46% of its revenue in July and August, and its ADR in August is 3.8 times its ADR in January	Apply dynamic pricing and minimum-stay rules through the summer, and run packages or discounts from November to February
The City Hotel cancels 41.7% of bookings and loses 10.89 million	Introduce controlled overbooking calibrated to the cancellation rate by lead time and segment
Non Refund bookings show a 99.4% cancellation rate	Audit how these bookings are recorded before deposit type is used in any pricing or policy decision

6.2 Limitations

- Revenue is estimated as ADR multiplied by nights; it is not taken from billing records, so the totals are indicative rather than exact.
- The file contains no booking identifier, so 31,994 identical rows could neither be confirmed as duplicates nor safely treated as separate bookings.
- July and August contain three years of arrivals while the remaining months contain two, which overstates the summer peak in Figure 4.
- The dashboard currently has no filter actions or parameters, so a reader cannot yet click a country or a segment and refilter the remaining views.
- The 99.4% cancellation rate on Non Refund bookings is almost certainly a recording effect, which limits what can be concluded about deposit policy from this data alone.

6.3 Future Work

- Add a "use as filter" action from the Regional Map and the Market Segment chart, and a hotel parameter so that the whole dashboard can be switched between the two properties.
- Add a year filter and a trend of cancellation rate by year, so that seasonality can be separated from growth.
- Build a predictive cancellation model from lead time, deposit type, market segment and country, and surface its risk score as an extra field in Tableau.
- Extend the analysis with occupancy and room-inventory data so that overbooking limits can be set numerically.

6.4 Summary of What Tableau Achieved

Tableau turned a file of 119,390 booking records into a single dashboard that answers all five objective questions at a glance. Calculated fields supplied the revenue and cancellation measures that the raw file lacked, so no external processing was needed. The combination of KPI cards, line charts, bar charts and a symbol map showed where revenue is earned, when it peaks and which groups of guests put it at risk, and the dashboard made those patterns visible together rather than one at a time. The result is a practical decision tool that a hotel revenue manager could use to set pricing, deposit rules and sales-channel strategy.